

Hangzhou DeepSeek Artificial Intelligence Basic Technology Research Co., Ltd., doing business as DeepSeek, is a Chinese artificial intelligence (AI) company that develops open weights large language models (LLMs). Based in Hangzhou, Zhejiang, DeepSeek is owned and funded by High-Flyer, a Chinese hedge fund. DeepSeek was founded in July 2023 by Liang Wenfeng, who serves as the CEO for both of the companies. The company launched an eponymous chatbot alongside its DeepSeek-R1 model in January 2025.

DeepSeek-R1 provided responses comparable to other contemporary LLMs, such as OpenAI's GPT-4 and o1. Its training cost was reported to be lower than other LLMs. The company claims that it trained V3 for US\$6 million, which was less than OpenAI's reported US\$100 million cost for GPT-4 in 2023. DeepSeek also claimed that they were using approximately one-tenth the computing power consumed by Meta's comparable model, Llama 3.1. DeepSeek's success against larger and more established rivals has been described as "upending AI".

DeepSeek's models are open-weight, meaning that the exact parameters are openly shared, but the training data is not openly licensed. Since the January 2025 debut of DeepSeek-R1, the company has made its new models available under free and open-source software licenses, like the MIT License. The company reportedly recruits AI researchers from top Chinese universities and hires from outside traditional computer science fields to broaden its models' knowledge and capabilities. DeepSeek is considered more closely linked to the defense industry of China, due to previous affiliations of its researchers, and its chatbot's non-combat role adoption by the People's Liberation Army since March 2025.

R1 was geopolitically significant as a open-weight, cost-effective, and high-performing release. The company also trained its models during ongoing US semiconductor trade restrictions on China, using a fewer number of weaker export model chips via a mixture of experts approach. Observers like the New York Post and The Guardian described R1 as a Sputnik moment in threatening the proprietary-led US AI ecosystem, including Nvidia, which lost US\$600 billion in market value, the largest single-company decline in US stock market history.

History

Pre-founding years

In June 2015, High-Flyer was co-founded by AI enthusiast Liang Wenfeng, who had been trading since the 2008 financial crisis while attending Zhejiang University. The company began stock trading using a GPU-dependent deep learning model on 21 October 2016. Previously, it had used CPU-based linear models. By the end of 2017, most of its trading was driven by AI.

Fire-flyer

In 2019, the company began constructing its first computing cluster, Fire-Flyer, at a cost of 200 million yuan. The computing cluster contained 1,100 GPUs interconnected at 200 Gbit/s and was retired after 1.5 years in operation. By 2021, Liang had started buying large quantities of Nvidia GPUs for an AI project, reportedly obtaining 10,000 Nvidia A100 GPUs before the United States restricted chip sales to China.

Fire-flyer 2

Computing cluster Fire-Flyer 2 began construction in 2021 with a budget of 1 billion yuan. It was reported that in 2022, Fire-Flyer 2's capacity had been used at over 96%, totaling 56.74 million GPU hours. 27% of Fire-Flyer 2's capacity was used to support scientific computing outside the company. Fire-Flyer 2 had 5,000 PCIe A100 GPUs in 625 nodes, each containing 8 GPUs. At the time, it exclusively used PCIe instead of the DGX version of A100. This was because the models it trained could fit within a single 40 GB GPU VRAM. Hence, there was no need for the higher bandwidth of DGX at the time (it required only data parallelism but not model parallelism). Later, it incorporated both NVLinks and Nvidia Collective Communications Library (NCCL) to train larger models that required model parallelism.

Founding

On 14 April 2023, High-Flyer announced the launch of an artificial general intelligence (AGI) research lab, stating that the new lab would focus on developing AI tools unrelated to the firm's financial business. Two months later, on 17 July 2023, that lab was spun off into an independent company. That lab was named DeepSeek, with High-Flyer as its principal investor and backer. Initially, venture capital investors were reluctant to provide funding, as they considered it unlikely that the venture would be able to quickly generate an exit (an ownership stake in an investment).

Company operation

DeepSeek is headquartered in Hangzhou, Zhejiang, and is owned and funded by High-Flyer. Its co-founder, Liang Wenfeng, serves as CEO. As of May 2024, Liang personally held an 84% stake in DeepSeek through two shell corporations. In February 2026, Anthropic accused DeepSeek of using thousands of fraudulent accounts to generate millions of conversations with Claude to train its own LLMs. In April 2026, investors began speaking with DeepSeek for a \$300 million funding round, which would bring DeepSeek to a total valuation of \$10 billion. In July 2026, Bloomberg and the Financial Times reported the company had begun preparations for an IPO that would see it list as soon as 2027.

Strategy

DeepSeek has stated that it focuses on research and does not have immediate plans for commercialization. This posture also means it can skirt certain provisions of China's AI regulations aimed at consumer-facing technologies.

DeepSeek's hiring approach emphasizes skills over lengthy work experience, resulting in many hires fresh out of university. The company likewise recruits individuals without computer science backgrounds to expand the range of expertise incorporated into the models, for

instance in poetry or advanced mathematics. According to The New York Times, dozens of DeepSeek researchers have or have previously had affiliations with People's Liberation Army laboratories and the Seven Sons of National Defence. Since 2025, its chatbot was adopted in non-combat roles by the People's Liberation Army, including hospitals, the People's Armed Police paramilitary, and national mobilization organizations.

Due to U.S. chip restrictions, DeepSeek has continuously refined its algorithms to maximize computational efficiency, leveraging older hardware and reducing energy consumption.

DeepSeek also expanded on the African continent as it offers more affordable and less power-hungry AI solutions. The company has bolstered African language models and generated a number of startups, for example in Nairobi. Along with Huawei's storage and cloud computing services, the impact on the tech scene in sub-Saharan Africa is considerable. DeepSeek offers local data sovereignty and more flexibility compared to Western AI platforms.

Training framework

High-Flyer/DeepSeek had operated at least two primary computing clusters: Fire-Flyer and Fire-Flyer 2. Fire-Flyer 1 was constructed in 2019 and was retired after 1.5 years of operation. Fire-Flyer 2 is still in operation as of 2025. Fire-Flyer 2 consists of co-designed software and hardware architecture. On the hardware side, Nvidia GPUs use 200 Gbps interconnects. The cluster is divided into two zones, and the platform supports cross-zone tasks. The network topology was two fat trees, chosen for high bisection bandwidth. On the software side, there are:

3FS (Fire-Flyer File System): A distributed parallel file system designed for asynchronous random reads. It uses Direct I/O and RDMA Read. In contrast to standard Buffered I/O, Direct I/O does not cache data. Caching is useless in this case, since each piece of data read is random and is not reused.

hreduce: Library for asynchronous communication, originally designed to replace Nvidia Collective Communication Library (NCCL). It is mainly used for allreduce, especially of gradients during backpropagation. It runs asynchronously on the CPU to avoid blocking kernels on the GPU. It uses two-tree broadcast like NCCL.

hfai.nn: Software library of commonly used operators for neural network training, similar to torch.nn in PyTorch.

HaiScale Distributed Data Parallel (DDP): Parallel training library that implements various forms of parallelism such as data parallelism (DP), pipeline parallelism (PP), tensor parallelism (TP), experts parallelism (EP), fully sharded data parallel (FSDP) and zero redundancy optimizer (ZeRO). It is similar to PyTorch DDP, which uses NCCL on the backend.

HAI Platform: Various applications such as task scheduling, fault handling, and disaster recovery.

As of 2022, Fire-Flyer 2 had 5,000 PCIe A100 GPUs in 625 nodes, each containing 8 GPUs. It later incorporated NVLinks and NCCL to train larger models that required model parallelism.

DeepSeek Harness

DeepSeek Harness is an open-source AI agent harness developed by DeepSeek. It provides an environment for building and running AI agents, with components including models, tools, skills, sessions, sandboxes, storage, scheduling, and user interfaces implemented as interchangeable plugins. The software uses the Cordis plugin architecture and is distributed under the MIT License. As of August 2026, DeepSeek Harness is available as a developer preview.

Model releases

DeepSeek Coder and DeepSeek LLM

DeepSeek released its first model, DeepSeek Coder, on 2 November 2023. The

DeepSeek-Coder V2 series was released shortly after. It was followed by the DeepSeek-LLM series on 29 November 2023.

Misc

In January 2024, DeepSeek released two DeepSeek-MoE models. On 3 April 2024, they released 3 DeepSeek-Math models.

V2 series

DeepSeek-V2 was released in May 2024. In September 2024, DeepSeek-V2.5 was introduced and revised in December.

V3 series

In December 2025, DeepSeek-V3-Base and DeepSeek-V3 (chat) were released. On 24 March 2025, DeepSeek released DeepSeek-V3-0324 under the MIT License, a revised version of V3.

On 21 August 2025, DeepSeek released DeepSeek V3.1 under the MIT License. This model features a hybrid architecture with thinking and non-thinking modes. It also surpasses prior models like V3 and R1, by over 40% on certain benchmarks like SWE-bench and Terminal-bench. It was updated to V3.1-Terminus on 22 September 2025, which fixed multiple language artifacts and improved in intelligence in general.

DeepSeek V3.2-Exp was released on 29 September 2025. It uses DeepSeek Sparse Attention, a more efficient attention mechanism based on previous research published in February. It was based on DeepSeek V3.1 Terminus. DeepSeek-V3.2 was released on 1 December 2025, alongside a DeepSeek-V3.2-Speciale variant that focused on reasoning.

R1 series

On 20 November 2024, the preview of DeepSeek-R1-Lite became available via chat. It was a

proprietary model, having only ever been released via API and web. On 20 January 2025, DeepSeek launched the DeepSeek-R1 model, which was free for iOS and Android. By 27 January, DeepSeek surpassed ChatGPT as the most downloaded freeware app on the iOS App Store in the United States, triggering an 18% drop in Nvidia's share price. DeepSeek subsequently released multiple distilled variants of DeepSeek R1, ranging from 1.5B parameters to 70B parameters.

On 28 May 2025, DeepSeek released DeepSeek-R1-0528 under the MIT License, which was a minor update upon R1. Its scores have also exceeded both DeepSeek R1 (January) and DeepSeek V3-0324 in certain benchmarks.

V4 series

On 24 April 2026, DeepSeek released a preview of its V4 series, including the 284-billion parameter DeepSeek-V4-Flash and the 1.6-trillion parameter DeepSeek-V4-Pro. Both feature a one million token context window, under the MIT License. V4 has been adopted by semiconductor manufacturers such as Huawei and Cambricon Technologies. The official versions of DeepSeek V4-Flash and V4-Pro were released on 31 July and 13 August, respectively.

List of models

The first DeepSeek models were essentially the same as Llama, which were dense decoder-only transformers. Later models incorporated the multi-head latent attention (MLA), MoE, and KV caching.

A decoder-only transformer consists of multiple identical decoder layers. Each of these layers feature two main components: an attention layer and a feedforward network (FFN) layer. V2 replaced the standard multi-head attention mechanism (MHA) with multi-head latent attention (MLA). This introduces compressed latent vectors to reduce key value (KV) cache size, and thus

memory usage.

A standard Mixture of Experts Transformer generally use the sparsely-gated MoE layers in the FFN layers. In such an MoE layer, there are several FFN modules in parallel (routed experts) and a small classifier (gate) to compute a score for all these modules upon each token. Only the highest-scoring modules are activated. Starting with DeepSeekMoE, DeepSeek adopted a variant that adds shared experts, which are always activated.

Technical specifications of models

DeepSeek's models are open-weight, which provides less freedom for modification than true open source software.

DeepSeek Coder

DeepSeek Coder is a series of eight models, four pretrained (Base) and four instruction-finetuned (Instruct). All have 16K context lengths. The model was made open-weights under the DeepSeek License, which includes restrictions like open and responsible downstream usage.

The training program was:

Pretraining: 1.8T tokens (87% source code, 10% code-related English (GitHub markdown and Stack Exchange), and 3% code-unrelated Chinese).

Long-context pretraining: 200B tokens. This extends the context length from 4K to 16K. This produced the Base models.

Supervised finetuning (SFT): 2B tokens of instruction data. This produced the Instruct models.

They were trained on clusters of A100 and H800 Nvidia GPUs, connected by InfiniBand, NVLink, NVSwitch.

DeepSeek-LLM

The DeepSeek-LLM series was released in November 2023. It has 7B and 67B parameters in both Base and Chat forms. DeepSeek's accompanying paper claimed benchmark results higher than Llama 2 and most open-source LLMs at the time. The model code is under the source-available DeepSeek License.

The architecture was essentially the same as the Llama series. Both had vocabulary size 102,400 (byte-level BPE) and context length of 4096. They trained on 2 trillion tokens of English and Chinese text obtained by deduplicating the Common Crawl.

The chat versions of the two base models were released concurrently, obtained by training base by supervised finetuning (SFT) followed by direct policy optimization (DPO). DPO is also known as reinforcement learning from human feedback.

DeepSeek-MoE

DeepSeek-MoE models (Base and Chat), each have 16B parameters (2.7B activated per token, 4K context length). The training was essentially the same as DeepSeek-LLM 7B, and was trained on a part of its training dataset. DeepSeek claimed performance comparable to a 16B MoE as a 7B non-MoE. It is a variant of the standard sparsely-gated MoE, with shared experts that are always queried, and routed experts that might not be. The company found this to help with expert balancing. In standard MoE, some experts can become overused, while others are rarely used, which wastes space. Attempting to balance expert usage causes experts to replicate the same capacity. They proposed that the shared experts to learn core capacities that are often used, and let the routed experts learn peripheral capacities that are rarely used.

DeepSeek-Math

DeepSeek-Math includes 3 models: Base, Instruct, and RL. Math was trained as follows:

Initialize with a previously pretrained DeepSeek-Coder Base v1.5 7B.

Further pretrain with 500B tokens (6% DeepSeekMath Corpus, 4% AlgebraicStack, 10% arXiv, 20% GitHub code, 10% Common Crawl). This produced Base.

Train an instruction-following model by SFT Base with 776K math problems and tool-use-integrated step-by-step solutions. This produced Instruct.

Reinforcement learning (RL) The reward model was a process reward model (PRM) trained from Base according to the Math-Shepherd method. This reward model was then used to train Instruct using group relative policy optimization (GRPO) on a dataset of 144K math questions related to GSM8K and MATH. The reward model was continuously updated during training to avoid reward hacking. This resulted in RL.

V2

In May 2024, DeepSeek released the DeepSeek-V2 series. The series includes 4 models, 2 base models (DeepSeek-V2, DeepSeek-V2 Lite) and 2 chatbots (chat variants). The two larger models were trained as follows:

Pretrain on a dataset of 8.1T tokens, using 12% more Chinese tokens than English ones.

Extend context length from 4K to 128K using YaRN. This resulted in DeepSeek-V2.

SFT with 1.2M instances for helpfulness and 0.3M for safety. This resulted in Chat SFT, which was not released.

RL using GRPO in two stages. The first stage was trained to solve math and coding problems. This stage used 1 reward model, trained on compiler feedback (for coding) and ground-truth labels (for math). The second stage was trained to be helpful, safe, and follow rules. This stage used 3 reward models. The helpfulness and safety reward models were trained on human

preference data. The rule-based reward model was manually programmed. All trained reward models were initialized from Chat (SFT). This resulted in the released version of Chat.

They opted for 2-staged RL, because they found that RL on reasoning data had unique characteristics different from RL on general data. For example, RL on reasoning could improve over more training steps.

The two V2-Lite models were smaller, and trained similarly. DeepSeek-V2 Lite-Chat underwent only SFT, not RL. They trained the Lite version to help further research and development on MLA and DeepSeekMoE.

Architecturally, the V2 models were significantly different from the DeepSeek LLM series. They changed the standard attention mechanism by a low-rank approximation called multi-head latent attention (MLA), and used the previously published MoE variant.

The Financial Times reported that it was cheaper than its peers with a price of 2 RMB for every million output tokens.

The DeepSeek-Coder V2 series included V2-Base, V2-Lite-Base, V2-Instruct, and V2-Lite-Instruct. Training process:

Base models were initialized from corresponding intermediate checkpoints after pretraining on 4.2T tokens (not the version at the end of pretraining), then pretrained further for 6T tokens, then context-extended to 128K context length.

DeepSeek-Coder and DeepSeek-Math were used to generate 20K code-related and 30K math-related instruction data, then combined with an instruction dataset of 300M tokens. This was used for SFT.

RL with GRPO. The reward for math problems was computed by comparing with the ground-truth label. The reward for code problems was generated by a reward model trained to predict whether a program would pass the unit tests.

DeepSeek-V2.5 was made by combining DeepSeek-V2-Chat and DeepSeek-Coder-V2-Instruct.

V3

DeepSeek-V3-Base and DeepSeek-V3 (chat variant) essentially uses the same architecture as V2 with the addition of multi-token prediction, which optionally decodes extra tokens faster but less accurately. Training process:

Pretraining on 14.8T tokens of a multilingual corpus, mostly English and Chinese. It contained a higher ratio of math and programming than the pretraining dataset of V2.

Extend context length twice, from 4K to 32K and then to 128K, using YaRN. This produced DeepSeek-V3-Base.

SFT for 2 epochs on 1.5M samples of reasoning (math, programming, logic) and non-reasoning (creative writing, roleplay, simple question answering) data. Reasoning data was generated by expert models. Non-reasoning data was generated by DeepSeek-V2.5 and checked by humans.

The expert models were trained by starting with an unspecified base model, then SFT on both <problem, original response> data, and synthetic <system prompt, prompt, problem, R1 response> data generated by an internal DeepSeek-R1-Lite model. The system prompt asked R1 to reflect and verify during thinking. Then the expert models were RL using an undisclosed reward function.

Each expert model was trained to generate just synthetic reasoning data in one specific domain (math, programming, logic).

Expert models were used instead of R1 itself, since the output from R1 itself suffered overthinking, poor formatting, and excessive length.

Model-based reward models were made by starting with a SFT checkpoint of V3, then finetuning on human preference data containing both final reward and chain-of-thought

leading to the final reward. The reward model produced reward signals for both questions with objective but free-form answers, and questions without objective answers (such as creative writing).

An SFT checkpoint of V3 was trained by GRPO using both reward models and rule-based reward. The rule-based reward was computed for math problems with a final answer (put in a box), and for programming problems by unit tests. This produced DeepSeek-V3.

DeepSeek released its DeepSeek-V3-0324 model, which used the same architecture as V3, on 24 March 2025 under the MIT License.

The DeepSeek team performed low-level engineering to improve efficiency, such as mixed-precision arithmetic. Much of the forward pass was performed in 8-bit floating point numbers (5E2M: 5-bit exponent and 2-bit mantissa) rather than the standard 32-bit, requiring special general matrix multiply (GEMM) routines to accumulate accurately. They used a custom 12-bit float (E5M6) only for the inputs to the linear layers after the attention modules. Optimizer states were in 16-bit (BF16). They minimized communication latency by extensively overlapping computation and communication, such as dedicating 20 streaming multiprocessors out of 132 per H800 for only inter-GPU communication. They lowered communication by rearranging (every 10 minutes) the exact machine each expert was on so as to avoid querying certain machines more often than others, adding auxiliary load-balancing losses to the training loss function, and other load-balancing techniques.

After training, it was deployed on clusters of H800 GPUs. The 8 H800 GPUs within a cluster were connected by NVLink, and the clusters were connected by InfiniBand.

The cost has been discussed and called misleading, because it covers only parts of the true cost.

Benchmark tests show that V3 outperformed Llama 3.1 and Qwen 2.5 while matching GPT-4o

and Claude 3.5 Sonnet.

R1

In January 2025, DeepSeek released the DeepSeek-R1 model under the MIT License.

DeepSeek-R1-Lite-Preview was trained for logical inference, mathematical reasoning, and real-time problem-solving. DeepSeek claimed that it exceeded performance of OpenAI o1 on benchmarks such as American Invitational Mathematics Examination (AIME) and MATH. However, The Wall Street Journal reported that on 15 problems from the 2024 edition of AIME, the o1 model reached a solution faster.

DeepSeek-R1 and DeepSeek-R1-Zero were initialized from DeepSeek-V3-Base and share its architecture. DeepSeek-R1-Distill models were instead initialized from other pretrained open-weight models, including LLaMA and Qwen, then fine-tuned on synthetic data generated by R1.

DeepSeek-R1-Zero was trained exclusively using GRPO RL without SFT. Unlike previous versions, it used no model-based reward. All reward functions were rule-based, mainly of two types (other types were not specified): accuracy rewards and format rewards. Accuracy reward was checking whether a boxed answer is correct (for math) or whether a code passes tests (for programming). Format reward was checking whether the model puts its thinking trace within a `<think>...</think>` tag.

R1-Zero has issues with readability and mixing languages. R1 was trained to address these issues and further improve reasoning:

SFT DeepSeek-V3-Base on thousands of cold-start data all with the standard format of `|special_token|<reasoning_process>|special_token|<summary>`, designed to improve model output readability.

Apply the same GRPO RL process as R1-Zero, adding a language consistency reward to encourage it to respond monolingually. This produced an unreleased internal model.

Synthesize 600K reasoning data from the internal model, with rejection sampling (i.e. if the generated reasoning had a wrong final answer, then it is removed). Synthesize 200K non-reasoning data (writing, factual QA, self-cognition, translation) using DeepSeek-V3.

SFT DeepSeek-V3-Base on the 800K synthetic data for 2 epochs.

Apply the same GRPO RL process as R1-Zero with rule-based reward (for reasoning tasks), but also model-based reward (for non-reasoning tasks, helpfulness, and harmlessness). This produced DeepSeek-R1.

Distilled models were trained by SFT on 800K data synthesized from DeepSeek-R1, in a similar way as step 3. They were not trained with RL.

V4

In April 2026, DeepSeek released a preview of their new V4 model series, also under the MIT License.

The DeepSeek V4 model improved on their previous V3/R1 architecture in the following ways:

The model uses their Manifold-constrained Hyper Connections (mHC) architecture, which is claimed by DeepSeek to enhance conventional residual connections.

They introduced Constrained Sparse Attention (CSA) and Heavily Compressed Attention (HCA), modifications to the Attention mechanism used within Transformer models based on their prior DeepSeek Sparse Attention architecture introduced in V3.2.

The Muon optimizer was used for most layers for faster convergence and improved training stability.

They released two sizes of model, V4-Flash and V4-Pro. Each are able to be operated in a non-reasoning mode, a reasoning mode, and a "Max" extended reasoning mode.

Legal status

In the United States, the National Defense Authorization Act for Fiscal Year 2026 instructs the United States Secretary of Defense and the Director of National Intelligence to remove and exclude any artificial intelligence developed by DeepSeek or owned by HighFlyer from devices operated by the United States Department of Defense and United States Intelligence Community or their contractors. Artificial intelligence by DeepSeek is only allowed with explicit permissions for research, training, and evaluation or military activities supporting national security functions such as counterterrorism or counterintelligence.

In Australia, the Department of Home Affairs issued a government-wide directive "to prevent the use or installation of DeepSeek products, applications and web services and where found remove all existing instances of DeepSeek products, applications and web services from all Australian Government systems and devices."

See also

Artificial intelligence industry in China

List of AI-assisted software development tools

List of LLMs

Lists of open-source artificial intelligence software

Reasoning model – Language models designed for reasoning tasks

Six Little Dragons – Informal group of Chinese tech startups

Zhejiang University – Public university in Hangzhou, Zhejiang, China

Notes

References

External links

[Official website](#)

[DeepSeek on GitHub](#)

[DeepSeek on Hugging Face](#)

[Official API documentation](#)

[Anthology of DeepSeek papers](#)

[Research blog of High-Flyer](#)